U.S.T.H.B. / Faculty of Mathematics — Academic year: 2026/2027
Department of Operations Research — Course: Thor (S1, bigDataA)

Exercise Sheet No. 2 (Scheduling on a Single Machine) — Solutions

Notation. C_i completion time, $L_i = C_i - d_i$ lateness (can be negative), $T_i = \max(0, L_i)$ tardiness, $U_i = 1$ if $C_i > d_i$ (task late) and 0 otherwise, $N_T = \sum U_i$ number of late tasks. All tasks are available at time 0 and the machine never stays idle. Every result below was checked by enumerating all possible orders.

Exercise 1: $1//\overline{C}$.

Rule used: SPT (Shortest Processing Time): sort the tasks by increasing p_i. SPT is optimal for $1//\overline{C}$.

Order	T_6	T_2	T_8	T_5	T_7	T_3	T_4	T_1
p_i	1	2	2	3	3	4	7	10
C_i	1	3	5	8	11	15	22	32

$\sum C_i = 97$, so $\overline{C} = 97/8 = \mathbf{12.125}$. Ties ($T_2, T_8$ and T_5, T_7) can be taken in either order: the value is the same.

Exercise 2: $1//L_{max}$.

Rule used: EDD (Earliest Due Date, Jackson's rule): sort by increasing d_i. EDD is optimal for $1//L_{max}$.

Order	T_2	T_6	T_4	T_7	T_3	T_1	T_5
d_i	20	30	48	50	67	80	100
C_i	13	16	36	45	76	105	112
L_i	-7	-14	-12	-5	9	25	12

$L_{max} = \mathbf{25}$ (task T_1).

Exercise 3:

1/ Sequence T_1, T_2, T_3: $C = (2, 3, 4)$, $L = (2-3,\ 3-6,\ 4-5) = (-1, -3, -1)$, so $L_{max} = \mathbf{-1}$.

2/ EDD gives T_1, T_3, T_2: $C = (2, 3, 4)$, $L = (-1, -2, -2)$, also $L_{max} = -1$. The sequence T_1, T_2, T_3 does not follow EDD but is optimal too (over the 6 orders the best value is -1, reached only by these two). Conclusion: EDD is a *sufficient* condition for optimality, not a necessary one, and the optimal schedule is not unique. Since $L_{max} < 0$, every task is finished before its due date.

Exercise 4:

1/ $T_{max} = \max(0, L_{max})$, so EDD also minimises T_{max}.

Order	T_1	T_4	T_2	T_3	T_7	T_6	T_5
d_i	8	24	42	44	68	85	90
C_i	6	16	34	46	62	79	89
L_i	-2	-8	-8	2	-6	-6	-1

$T_{max} = \mathbf{2}$ (task T_3), with the order $T_1, T_4, T_2, T_3, T_7, T_6, T_5$.

2/ *Rule used:* Moore–Hodgson: go through the EDD order; as soon as a task finishes late, remove the longest task scheduled so far and put it at the end.

EDD: T_1 ends at 6, T_4 at 16, T_2 at 34, T_3 at 46 $>$ 44: late. The longest task among $\{T_1, T_4, T_2, T_3\}$ is T_2 ($p = 18$): remove it. The time goes back to 28, then T_7 ends at $44 \leq 68$, T_6 at $61 \leq 85$, T_5 at $71 \leq 90$.

Order	T_1	T_4	T_3	T_7	T_6	T_5	T_2
C_i	6	16	28	44	61	71	89
d_i	8	24	44	68	85	90	42
U_i	0	0	0	0	0	0	1

One late task (T_2): $\sum U_i = \mathbf{1}$, i.e. $\overline{U} = 1/7$ if $\overline{U}$ is taken as a mean. Enumeration confirms that no order puts all 7 tasks on time.

Exercise 5: $1//N_T$ (Moore–Hodgson, as in Exercise 4.2).
EDD order: $T_9, T_4, T_5, T_6, T_{10}, T_2, T_1, T_8, T_3, T_7$ with completions $1, 3, 7, 11, 15, 18, \ldots$ The first late task is T_2: $C = 18 > 16$. The longest tasks scheduled so far are T_5, T_6, T_{10} ($p = 4$, a tie): remove one of them, e.g. T_5. The time goes back to 14, and all the remaining tasks are on time:

Order	T_9	T_4	T_6	T_{10}	T_2	T_1	T_8	T_3	T_7	T_5
C_i	1	3	7	11	14	19	20	21	23	27
d_i	2	3	14	15	16	19	23	25	31	8
U_i	0	0	0	0	0	0	0	0	0	1

$N_T = \mathbf{1}$ (removing T_6 or T_{10} instead also gives $N_T = 1$). It cannot be 0: EDD itself has a late task, and EDD puts every task on time whenever this is possible.

Exercise 6: **Model.** Paints are tasks, cleaning times are sequence-dependent setup times c_{ij}. This is 1/set-up/C_{max}:

$$C_{max} = \sum p_i + \sum(\text{cleanings between consecutive colours}) = 73 + \sum c_{ij}.$$

Since $\sum p_i = 13 + 14 + 15 + 16 + 15 = 73$ h is fixed, we must find a Hamiltonian path of minimum length in the complete directed graph whose vertices are the colours and whose arcs are weighted by c_{ij}.

Complexity. This is a travelling-salesman-type problem, NP-hard in general. Here, with 5 colours, there are only $5! = 120$ orders, so enumeration is possible. In general we use the heuristic from the course (start with the cheapest arc, then always go to the cheapest colour not yet produced).

Solution. The cheapest arcs (cost 1) are White→Yellow, Yellow→Red, Red→Violet, Violet→Blue. Starting with White→Yellow, the heuristic follows exactly this chain:

$$\text{White} \to \text{Yellow} \to \text{Red} \to \text{Violet} \to \text{Blue}, \qquad \sum c_{ij} = 1 + 1 + 1 + 1 = 4.$$

This is optimal: any order needs 4 changes and every change costs at least 1, so 4 is a lower bound. Total production time $= 73 + 4 = \mathbf{77}$ hours (light to dark colours). Note that the heuristic depends on the tie: starting with Red→Violet it gives 18 h of cleaning (91 h in total).

Exercise 7:

1/ Mean flow time $= \overline{C}$ (all tasks are available at 0): SPT.

Order	7	6	3	1	2	4	5
p_i	1	2	3	4	5	6	7
C_i	1	3	6	10	15	21	28

$\sum C_i = 84$, so $\overline{C} = 84/7 = \mathbf{12}$ days.

2/ Largest lateness L_{max}: EDD (tasks 2 and 4 both have $d = 10$, either order works).

Order	1	2	4	7	3	5	6
d_i	8	10	10	14	15	17	20
C_i	4	9	15	16	19	26	28
L_i	-4	-1	5	2	4	9	8

$L_{max} = \mathbf{9}$ days (task 5).

Exercise 8: Lateness $L_i = C_i - d_i$ (negative means early). Times in minutes.

1/ Order 3, 4, 1, 5, 2:

Product	3	4	1	5	2
Start	0	80	280	330	360
C_i	80	280	330	360	510

$\overline{C} = 1560/5 = \mathbf{312}$ min.

2/ SPT is optimal: order 5, 1, 3, 2, 4 with $C = (30, 80, 160, 310, 510)$, so $\overline{C} = 1090/5 = \mathbf{218}$ min.

3/ Order 5, 1, 3, 2, 4 (the SPT order):

Product	5	1	3	2	4
C_i	30	80	160	310	510
d_i	200	100	410	300	400
L_i	-170	-20	-250	10	110

$L_{min} = \mathbf{-250}$, $L_{max} = \mathbf{110}$, $\overline{L} = -320/5 = \mathbf{-64}$ min.

4/ EDD is optimal for L_{max}: order 1, 5, 2, 4, 3.

Product	1	5	2	4	3
C_i	50	80	230	430	510
d_i	100	200	300	400	410
L_i	-50	-120	-70	30	100

$L_{min} = \mathbf{-120}$, $L_{max} = \mathbf{100}$, $\overline{L} = -110/5 = \mathbf{-22}$ min.

5/ Order 1, 5, 2, 3, 4: $C = (50, 80, 230, 310, 510)$, $L = (-50, -120, -70, -100, 110)$, so $\overline{L} = -230/5 = \mathbf{-46}$ min (and $L_{max} = 110$).

Conclusion. $\overline{L} = \overline{C} - \overline{d}$ and $\overline{d} = 1410/5 = 282$ is fixed, so minimising the mean lateness is the same as minimising $\overline{C}$: SPT gives the best value ($218 - 282 = -64$). Order 1, 5, 2, 3, 4 (-46) is worse than SPT, and EDD (-22) is the worst of the three. EDD minimises the *largest* lateness, not the mean one: each criterion needs its own rule. (In terms of tardiness, $T_i = \max(0, L_i)$, the order 1, 5, 2, 3, 4 has a single late product and a mean tardiness of 22 min, the best of all 120 orders.)